PARISHRAM 2.0

CLASS 12TH 2026

Botany-Sexual Reproduction in Flowering Plants

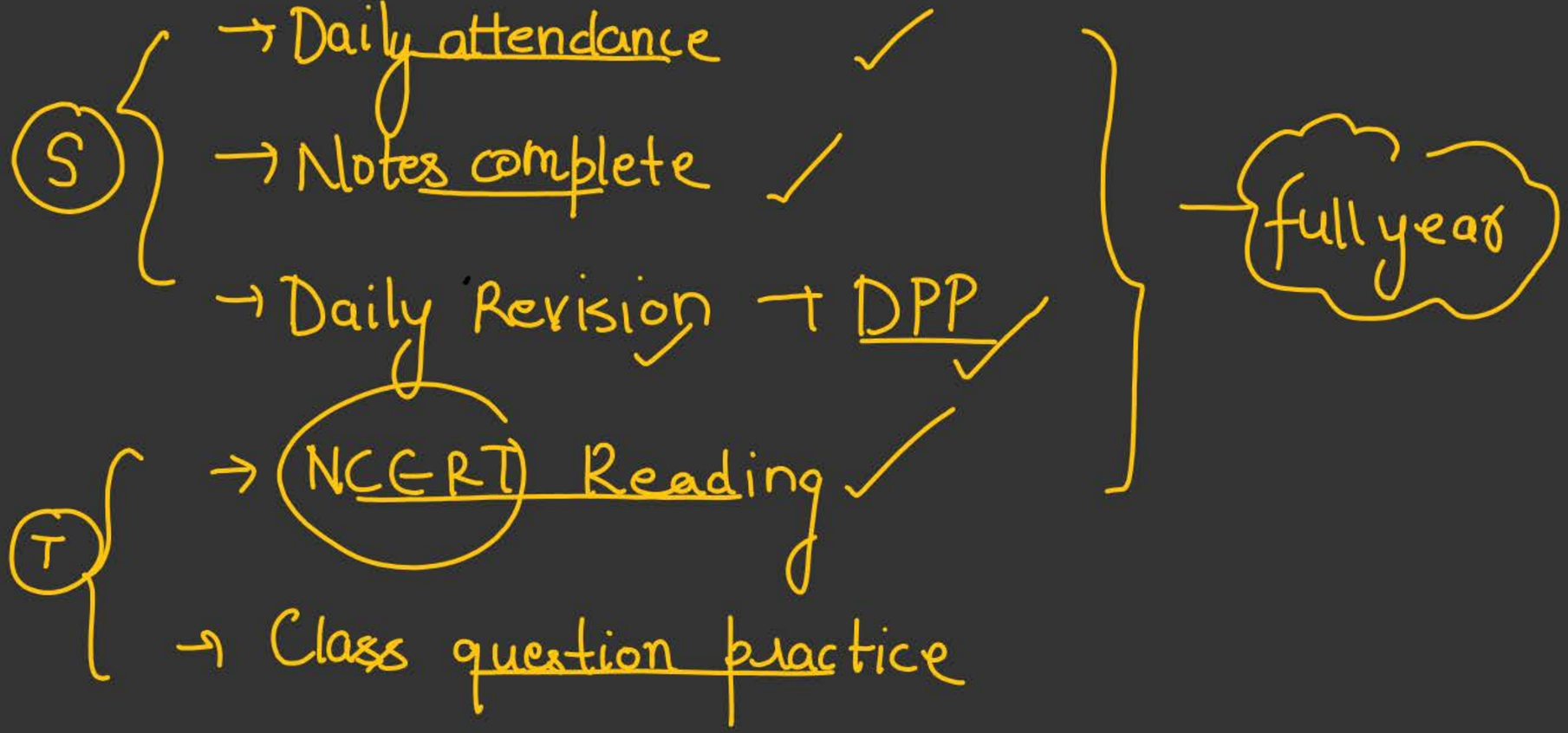
BIOLOGY LECTURE-1

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TOPICS TO BE COVERED

1. Flower
2. Structure of stamen
3. Microsporogenesis
4. Type option here


S
→ Daily attendance ✓
→ Notes complete ✓
→ Daily Revision + DPP ✓
T
→ NCERT Reading ✓
→ Class question practice
full year

Flower

externally fundamentally

To a biologist, flowers are morphological and embryological marvels and the sites of sexual reproduction. In earlier classes, you have read the various parts of a flower. Figure 1.1 will help you recall the parts of a typical flower. Can you name the two parts in a flower in which the two most important units of sexual reproduction develop?

→ Androecium & Gynoecium.

Sexual Reproduction in Flowering plants

is a biological phenc
where parents give
their young ones

Angiosperms

most evolved group of
plant kingdom

Sexual Reproduction

(complex, elaborate & slow process)

- Pre-fertilization event
- Fertilization
- Post-fertilization

Most important requirement →

1. Formation of gamete
2. Transfer of male gamete to the female gamete [Pollination]

→ Pre fertilization

3. Fusion of gamete = syngamy → Fertilization

④ → Formation of zygote

⑤ → Formation of embryo

] → Post-fertilization change

— α —

In Angiosperms, Flowers are the sex organ responsible for Sexual reproduction

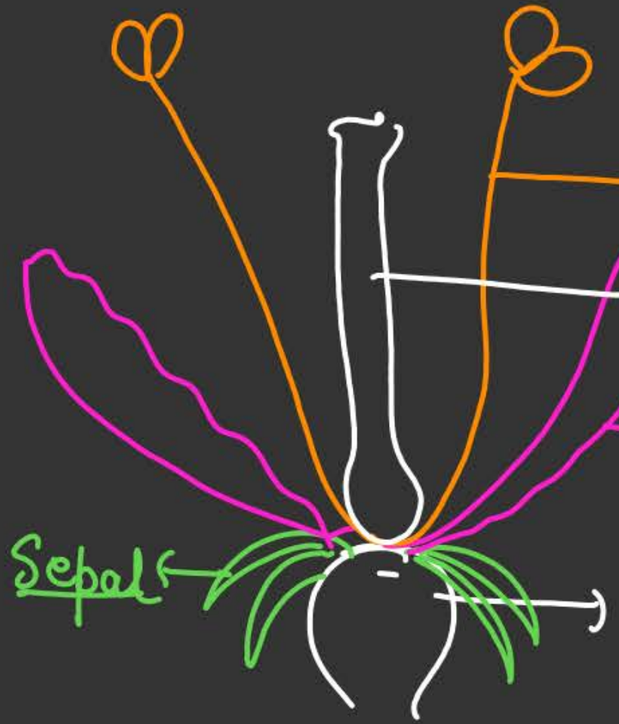
However for humans, flower is just a pleasure aesthetic

Structure of flower.

→ 4 parts

- Essential [ज़रूरी]
  - Androecium [Stamen] → ♂ sex organ = ♂ gamete
  - Gynoecium (Carpel.) → ♀ sex organ = ♀ gamete
- Non essential [not so necessary]
  - Calyx [Sepal]
  - Corolla [petal]

These two whorls are not directly involved in reproduction



nus / torus / receptacle

Structure of Androecium [collective term for stamen]

→ Stamen →



→ Anther — main site for the formation of pollen grain.

→ Connective [holds the 2 lobes of anther]

→ Filament (Thin, long slender stalk)

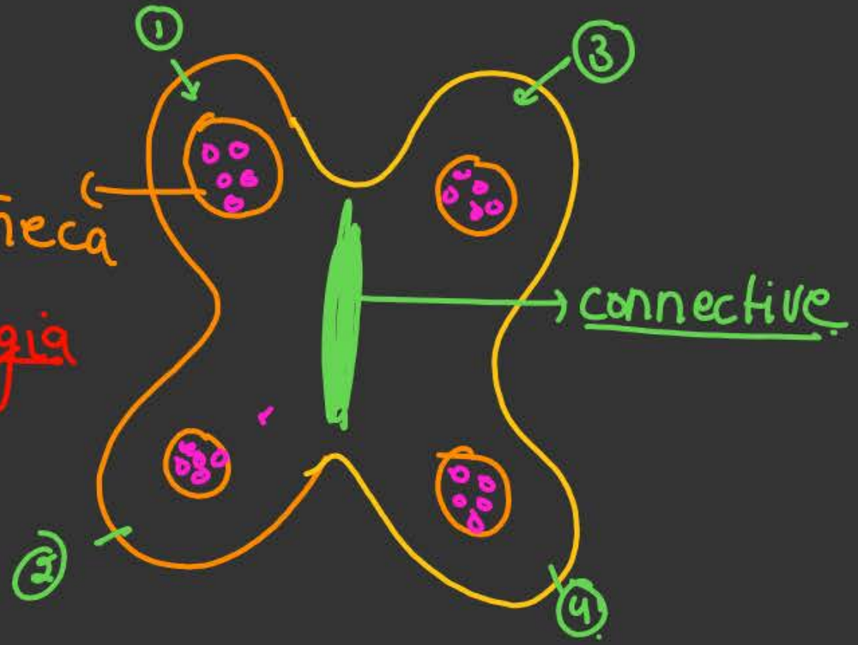


→ Proximal end

thalamus | petal

Theca

Sporangia



Anther → Bilobed

→ Tetragonal [4-sided]

- Tetrathecous

[but in each lobe
= 2 theca {dithecous}]

- Tetrasporangiate

1.2.1 Stamen, Microsporangium and Pollen Grain

Figure 1.2a shows the two parts of a typical **stamen** – the long and slender stalk called the **filament**, and the terminal generally bilobed structure called the **anther**. The proximal end of the filament is attached to the thalamus or the petal of the flower. The number and length of stamens are variable in flowers of different species. If you were to collect a stamen each from ten flowers (each from different species) and arrange them on a slide, you would be able to appreciate the large variation in size seen in nature. Careful observation of each stamen under a dissecting microscope and making neat diagrams would elucidate the range in shape and attachment of anthers in different flowers.

A typical angiosperm anther is **bilobed** with each lobe having two theca, i.e., they are **dithecous** (Figure 1.2b). Often a longitudinal groove runs lengthwise separating the theca. Let us understand the various types of tissues and their organisation in the transverse section of an anther (Figure 1.3a). The bilobed nature of an anther is very distinct in the transverse section of the anther. The anther is a four-sided (tetragonal) structure consisting of four **microsporangia** located at the corners, two in each lobe.

The microsporangia develop further and become **pollen sacs.** They extend longitudinally all through the length of an anther and are packed with pollen grains.

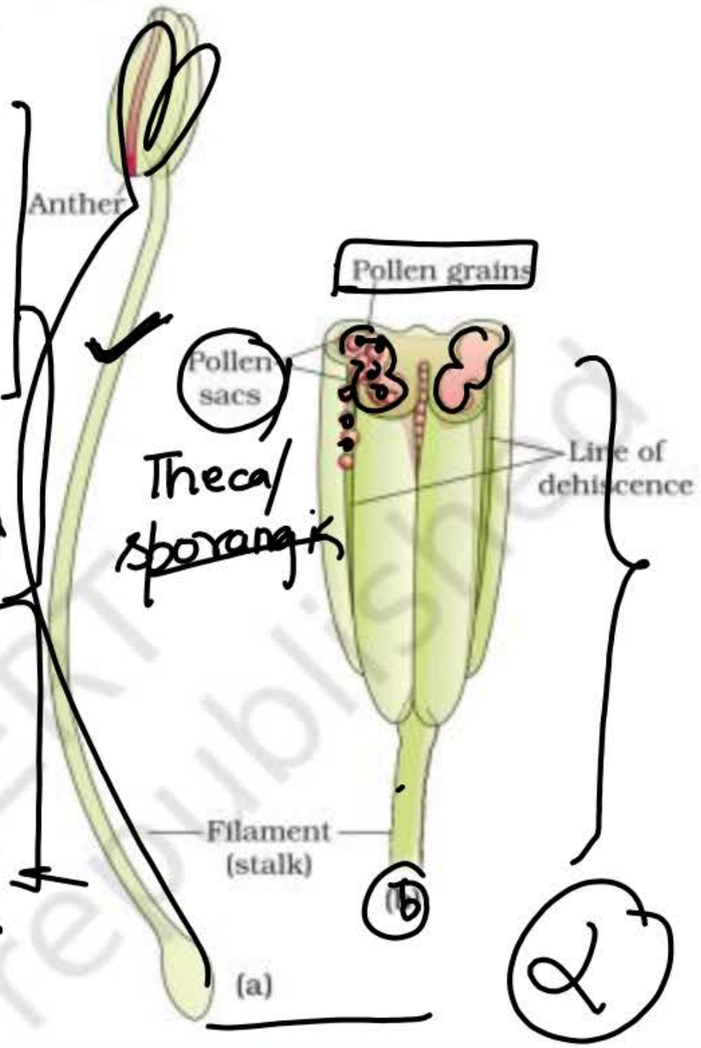


Figure 1.2 (a) A typical stamen; (b) three-dimensional cut section

Homework

Type Here → Complete notes

→ Read NCERT.

THANK YOU